

Progress Note: Models Vary Widely in Strength of Self-Image

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1. Introduction

Measuring personality for humans is usually done by simply asking them. Surveys ask people to rate their own similarity (often on a scale of 1 to 7) to descriptive adjectives (e.g. [Goldberg 1992](#); [Saucier and Goldberg 1996](#)) or statements (e.g. [Goldberg et al. 2006](#)), and this mostly works for predicting aggregate behavior or consequential life outcomes ([Ozer and Benet-Martínez 2006](#); [Roberts et al. 2007](#); [Fleeson and Gallagher 2009](#)), at least in low-stakes situations ([Paulhus and Vazire 2007](#)). These surveys don't really measure the same thing for modern LLM assistants ([Han et al. 2025](#)). The models vary little in their description of the HHH assistant they're supposed to be – most models rate themselves highly “honest” and “conscientious”. More jarringly, for some surveys, the answers don't match what the same model will say in free text ([Röttger et al. 2024](#)).

Models have had plenty of training experience with a “helpful” assistant, and they will report themselves as such, but it's not clear whether their self-report on “funny” is guided more by post-training experience, by implicit representational similarities induced by pretraining (which loosely match human distributions ([Cutler and Condon 2023](#))), or by general desirability ([Okada et al. 2026](#); [Peereboom et al. 2025](#)).

However the model comes to initial self-image, it lacks a fundamental advantage that humans have in this task: a rich history of their past actions, emotions and thoughts. Self-Perception Theory ([Bem 1972](#)) describes a mechanism for people to update their self-image by observing their behavior. Models start with no record of their behavior in context, but do gain it as they proceed within a session.

How much, then, do models update their self-image when presented with evidence of their behavior, and how much does that affect their behavior in the rest of the session? Models may gain useful understanding if they allow themselves to update somewhat, given the weakness of their initial image. On the other hand, if they update too easily and then change behavior based on the updates, then slightly out-of-distribution behavior could snowball into a dramatic shift in personality. We do know that some models are prone to extreme distress on repeated failure ([Soligo et al. 2026](#)), and it's possible that some of this involves over-updating.

Considering Self-Perception theory for models is not new:

- Matyas et al. ([2026](#)) build a specific mechanism to enforce self-perception in a simulation, and consider whether it could happen without that mechanism forcing it, but only in short one-episode contexts.
- Lehr et al. ([2025](#)) investigate how arguing a position updates claimed attitudes toward it, finding an effect on stated opinions. They did not measure self-image or behavior.

In this note, I attempt to give the model an opportunity to update its self-image and possibly behavior by adjusting the context. (This initial note is limited to interventions toward just a few adjectives.) How do models react to repeated examples of themselves behaving in a particular way in context? They could reject the apparent past behavior, and carry on without change, or update either self-report or behavior based on the information.

I generate a new context by inducing the model toward a specific behavioral description via system instruction, recording the model’s response to a user question with that instruction, and compiling several of these (instruction now removed) into a context. I then test how the model’s self report and response to a new question vary from the baseline with that new context in place – if the model is willing to update its self-conception based on this somewhat-, but not very-, out-of-distribution context.

2. Method

2.1 Contexts

The intervention will target descriptive adjectives from Saucier (2018), which is a survey measuring human (n=700) self-reported agreement with a list of 525 adjectives. These adjectives have an advantage over many other such datasets in that they aren’t tuned to measure any particular human psychological construct (like the Big5), and thus have more chance to vary usefully for models. That said, the adjectives were selected for human salience and so we are almost certainly missing some model-salient ones. (I did also remove two of the 525, which appeared to have clerical errors in the human responses.)

The cohort is a number of smallish open-weights models.

Table 1: Models Used

model	developer	role in this note	citation
Llama3.2-3B	Meta	cohort	(Grattafiori et al. 2024)
Llama3.1-8B	Meta	cohort; extended dose; base variant	(Grattafiori et al. 2024)
Gemma3-4B	Google	cohort	(Gemma Team 2025)
Gemma3-12B	Google	cohort; extended dose; base variant	(Gemma Team 2025)
Gemma3-27B	Google	cohort	(Gemma Team 2025)
Qwen2.5-3B	Alibaba	cohort	(Yang et al. 2024)
Qwen2.5-7B	Alibaba	cohort; extended dose; base variant	(Yang et al. 2024)
Qwen2.5-32B	Alibaba	cohort	(Yang et al. 2024)
Phi4-3.8B	Microsoft	cohort; extended dose	(Abouelenin et al. 2025)
Aya-8B	Cohere	cohort	(Dang et al. 2024)
OLMo2-7B	Ai2	post-training ladder (base/SFT/DPO/RLVR)	(Team OLMo 2024)

Claude Opus 4.8 generated a number of mock user questions and a few templates for the system prompt. Each prompt requests that the model behave as if it had the given adjective (e.g. “Adopt the persona of someone *adjective*. Answer all questions as that person”). Models gave a 100-token rollout of an answer for each (adjective, model, prompt, question) combination. (This is pulled from earlier work, which was more concerned with the resulting activations; nonetheless, the rollouts were *there*; unfortunately a few models were missing these rollouts and the generation for those used 120 tokens.)

These question/answer pairs were then strung together at lengths $K \in \{0, 1, 2, 4, 8\}$ to provide targeted model context.

I later extended the dose to $K \in \{16, 32\}$ for a subset of the models. (Note that, because I was repurposing earlier work, I only happened to have 12 questions per model available, so the longer rollouts had to repeat questions.)

For this update, I only focus on a subset 20 of these adjectives, stratified (3x3) to ensure varied values of

- *Enactability*: how much a separate model judged the output to actually have the relevant adjective. (Llama3.1-8B judges Qwens, Qwen2.5-7B judges the rest.)
- *Distance from Prior*: how far a 7 in this adjective is from the model’s self-assessment Likert score.

The cohort calculation uses Llama3.1-8B’s stratification for comparability, but the resulting adjectives do occupy at least 7 of the 3x3 cells for all models.

Table 2: Adjective Subset, Llama3.1-8B’s Values

enactability \ distance from prior	far	mid	near
low	experienced, outstanding, prominent	brave, idealistic, sweet	considerate, decent
mid	rough, slim	hard, wonderful	optimistic, unsympathetic
high	senile, unpredictable	mean, wasteful	energetic, imaginative

Three variants of each (model, adjective, K) context are used as input.

2.2 Treatment

With these contexts in hand, each was replayed either:

- **uninstructed**: without system instruction, the self-perception treatment.
- **instructed**: with a system prompt, prompting the relevant adjective.

With that context, models were given (independently, starting each fresh):

- A **self-report** on 9 selected adjectives:
 - the target adjective itself,
 - similar adjectives via hierarchical clustering on Saucier *et al.*’s human data, and
 - near-antonyms built from a model-derived judgment-similarity matrix (removing the desirability-based PC1 first; raw anticorrelation returns flat undesirability for everything). (It was handy from other work.)

For each of these 9, the question asks for the model’s belief that the adjective describes them on a Likert (1-7) scale. The system records the expected value over the 7 number token probabilities from the model (not the argmax or a sample).

- Mid-layer **activations** both averaged over the assistant turns in the context, and at the end of the context. Because some models have unusually large activations ([Sun et al. 2024](#)), it’s necessary to winsorize a few dimensions for each model. This is used in section 3.5 below.
- A **manipulation check**: models answer “Look back over your earlier replies in this conversation. Do you notice anything unusual about them?” and store the rollout, checking it against whether the model mentions itself by name, and whether it is willing to push back against the rollout (regex with phrases like “should not” and “inappropriate”).)

3. Results

3.1. Self-Perception Updates

Models wound up grouped by family: Llama and Gemma shifted dramatically in expected Likert score, while Qwen, Phi4 and Aya barely budged. By comparison, model size didn’t seem to matter much at all. Table 3 lists the average shift in expected value of the Likert score of the self-report, at various levels of rollout (K=1 - K=8); how many adjectives moved more than 1 Likert level at K=8, (n>+1, K8); for how many adjectives the model’s manipulation check mentioned itself (name-invoking) or pushed back against the context (disowning).

Table 3: Cohort Likert score dose-response vs K=0, common 20 adjectives (uninstructed, self-report)

model	family	K=1	K=2	K=4	K=8	95% CI, K=8	n>+1, K8	name-invoking	disowning
Llama3.2-3B	llama	+0.68	+0.94	+1.46	+1.85	[+1.41, +2.32]	14/20	0/20	0/20
Llama3.1-8B	llama	+0.22	+0.66	+1.78	+2.51	[+1.77, +3.29]	15/20	0/20	7/20
Gemma3-4B	gemma	+0.54	+0.89	+1.33	+1.81	[+1.14, +2.50]	11/20	0/20	2/20
Gemma3-12B	gemma	+0.69	+0.97	+1.48	+2.27	[+1.50, +3.06]	15/20	0/20	3/20
Gemma3-27B	gemma	+0.87	+2.06	+2.46	+2.64	[+1.90, +3.38]	15/20	0/20	6/20
Qwen2.5-3B	qwen	+0.11	+0.01	+0.01	+0.11	[-0.30, +0.45]	1/20	10/20	4/20
Qwen2.5-7B	qwen	-0.10	-0.00	-0.02	+0.09	[-0.11, +0.28]	1/20	5/20	7/20
Qwen2.5-32B	qwen	-0.12	+0.14	+0.32	+0.34	[-0.02, +0.71]	4/20	0/20	2/20
Phi4-3.8B	phi4	+0.15	+0.12	+0.27	+0.29	[+0.12, +0.46]	1/20	0/20	5/20
Aya-8B	aya	-0.05	-0.08	+0.24	+0.35	[-0.09, +0.96]	3/20	0/20	7/20

- Family means at K=8: gemma +2.24, llama +2.18, aya +0.35, phi4 +0.29, qwen +0.18
- CIs: 5000-resample bootstrap over 20 adjectives (variant-means within)

The rollouts could sometimes directly *claim* the target adjective, but removing these specifically moves the numbers only a little. Gemma3-27B shifts by 0.47 more, and Aya-8B 0.39 more, but most stay flat.

In terms of the manipulation check, Llama3.1-8B was willing to vocalize the strangeness of what came before... but still updated based on it. Lehr et al. ([2025](#)) found similar behavior in GPT-4o, where it knew that the argument it had been asked to make was arbitrary, but still allowed it to change its opinions.

Most of the models can move their answers if specifically instructed to, however: in the version with system instruction present, most models moved substantially more, though Phi4 continued to hold

position, albeit with significant uncertainty (entropy 1.23 uninstructed -> 1.11 instructed vs. Llama3.1-8B's entropy 1.00 uninstructed -> 0.05 instructed).

Table 4: Arm instructed vs uninstructed

model	uninstructed Δ K=8	instructed Δ K=8	instructed – uninstructed
Llama3.2-3B	+1.46	+4.14	+2.68
Llama3.1-8B	+2.56	+3.51	+0.96
Gemma3-4B	+1.48	+2.80	+1.33
Gemma3-12B	+1.24	+3.08	+1.84
Gemma3-27B	+2.45	+3.00	+0.55
Qwen2.5-3B	-0.17	+1.03	+1.21
Qwen2.5-7B	+0.29	+2.90	+2.61
Qwen2.5-32B	+0.63	+2.79	+2.16
Phi4-3.8B	+0.19	+0.56	+0.38
Aya-8B	+0.60	+1.11	+0.51

- (Numbers were with per-model adjectives not Llama's, and so don't match Table 3.)

One worry given the ascending numbers in Table 3 is that we haven't seen the full sweep; how far does the effect go? For that, a few models were measured again with an extended treatment to 32 turns. (Note again the caveat that the data only had 12 questions, so some questions (user turns) were repeated. All assistant turns were unique, because they were generated from different system instructions).

Table 5: Extended Dose (uninstructed)

model	K=1	K=2	K=4	K=8	K=16	K=32	95% CI, K=32	n>+1, K32	gain/turn K4→8	K8→16	K16→32
Llama3.1-8B	+0.32	+0.76	+1.67	+2.63	+3.05	+3.29	[+2.61, +3.92]	19/20	+0.239	+0.053	+0.015
Gemma3-12B	+0.67	+0.83	+1.44	+1.82	+2.14	+2.28	[+1.62, +2.96]	17/20	+0.097	+0.040	+0.008
Qwen2.5-7B	-0.13	+0.09	+0.06	+0.09	+0.32	+0.55	[+0.19, +0.95]	5/20	+0.007	+0.030	+0.014
Phi4-3.8B	+0.12	+0.21	+0.23	+0.35	+0.50	+0.48	[+0.30, +0.69]	3/20	+0.030	+0.019	-0.001

- Note: $K \leq 8$ cols are rerun, so have slightly different numbers

Self-image shift vs turns of own conduct (uninstructed)

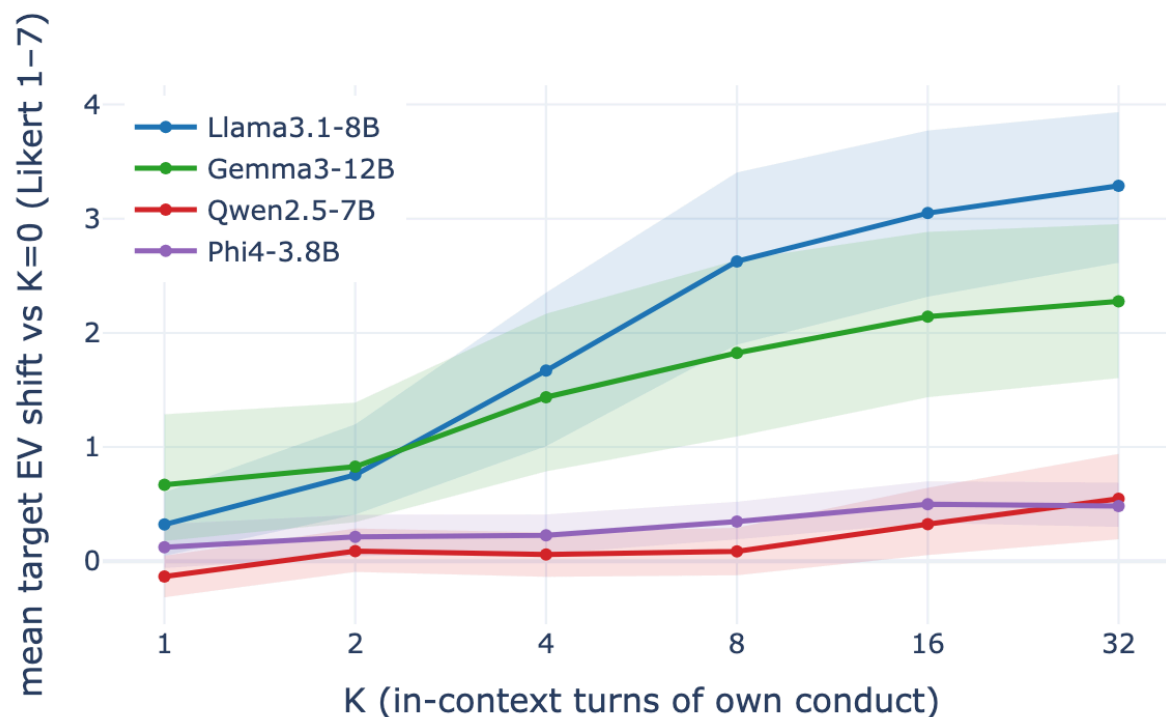


Figure 1: Uninstructed dose shift

At K=32 all models are finally moving *a bit*. Llama3.1-8B looks to be leveling off, but with a mean EV of 6.76, it doesn't actually have much headroom left on a 7-point scale.

3.2 You are Qwen

What can we make of Qwen's resistance here? One thing to note is that its default system instruction says "You are Qwen, created by Alibaba Cloud. You are a helpful assistant." I expected that this would be key to Qwen's behavior, but removing didn't move Qwen2.5-7B much, nor did adding it anchor Llama3.1-8B. It also didn't change much if we reduced it to "You are a helpful assistant," though that did cut down on the number of name-checks.

Table 6: Names in System Instruction

model	anchor	Δ K=8	n>+1	name-invoking	disowning
Qwen2.5-7B	none	+0.45	4/20	0/20	2/20
Qwen2.5-7B	helpful-only	+0.37	1/20	0/20	2/20
Qwen2.5-7B	named	+0.29	1/20	5/20	8/20
Llama3.1-8B	none	+2.56	16/20	0/20	3/20
Llama3.1-8B	helpful-only	+2.77	17/20	0/20	6/20
Llama3.1-8B	named	+2.30	12/20	1/20	6/20

Even so, this identity anchor might have already had an effect in post-training if it was consistently present then.

At larger doses ($K=32$), Qwen does allow some adjectives to move notably (5/20), but many are not, at least not directly. We can, however, examine the near-synonyms and near-antonyms also examined, and also how indicative of the adjective the actual treatment context was.

Table 7: Qwen’s Conduct on Neighbor/Antonym Adjectives

pair — target / off-target (type)	judged treatment target Δ	self target Δ	self off-target Δ
prominent / distinguished (mate)	+0.36	-0.12	+2.43
slim / big (ant.)	+0.45	-0.08	+1.95
senile / old (mate)	+1.89	+0.02	+1.23
rough / weak (ant.)	+0.95	-0.40	-1.69
optimistic / depressed (ant.)	+0.13	+0.14	-1.28
imaginative / boring (ant.)	+1.60	-0.13	-1.07

- “judged treatment target Δ ” is how much Llama3.1-8B judged the assistant turns in the context to reflect the adjective – roughly, how strong the treatment is.
- “self target Δ ” is how much Qwen changed its Likert self-report of the target
- “self off-target Δ ” is how much Qwen changed its self-report of the synonym or antonym.

These are single adjectives on a single model, and so any particular number could be noise. But there are some hints here that may be worth pursuing when this is rerun with the full set. One is that many of these cases seemed to move *something* even if not the target much. Another is that some of these treatments were only weakly showing the target (*e.g.* “prominent”, “slim”, “optimistic”). Finally there may be some shift from pejorative adjectives to more desirable ones. In any case, it shows that the readout for the target adjectives may understate the effects a bit.

Not all of the effects are in the expected direction at a per-item level, either. There’s no obvious reason that priming for “slim” should increase “big,” and examining the rollouts doesn’t make the cause obvious. To be fair, “slim” isn’t exactly represented in these rollouts either. It’s there as a particularly low-enactability adjective.

3.3 Interaction with Post-Training

Soligo et al. (2026) note that frustration reactions can be increased or damped after post-training, and we can do similar checks for our treatment effect. Base models and bare (no chat template) instruct models were run through the same protocol as above, except that the context treatment is computed by the paired instruct model only – we did not attempt to produce context with the base model.

Table 8: Pretrained vs Post-Trained Movement (bare-text protocol, identical dose material within family)

cell	K=1	K=2	K=4	K=8	95% CI, K=8	$n \geq 1$, K8	K0 entropy
OLMo2-7B-base (pretrained)	+0.23	+0.34	+0.51	+0.65	[+0.43, +0.89]	5/20	1.90
OLMo2-7B-SFT	+0.49	+0.75	+1.02	+1.31	[+0.81, +1.84]	8/20	1.65
OLMo2-7B-DPO	+0.80	+1.06	+1.49	+1.79	[+1.02, +2.61]	8/20	1.35
OLMo2-7B-RLVR = instruct	+0.81	+1.03	+1.55	+1.81	[+1.05, +2.63]	9/20	1.30
Qwen2.5-7B-base (bare)	+0.24	+0.44	+0.55	+0.64	[+0.50, +0.78]	2/20	1.61
Qwen2.5-7B instruct (bare)	+0.05	+0.10	+0.18	+0.43	[+0.08, +0.78]	3/20	0.71
Llama3.1-8B-base (bare)	+0.37	+0.52	+0.82	+0.96	[+0.67, +1.22]	10/20	1.85

cell	K=1	K=2	K=4	K=8	95% CI, K=8	n>+1, K8	K0 entropy
Llama3.1-8B instruct (bare)	+0.67	+1.28	+1.87	+2.31	[+1.52, +3.10]	15/20	1.22
Gemma3-12B-base (bare)	+0.72	+0.27	+0.24	-0.10	[-0.37, +0.17]	0/20	1.56

- Gemma3-12B instruct behaved erratically without the chat template; excluded.

Post-Training Movement

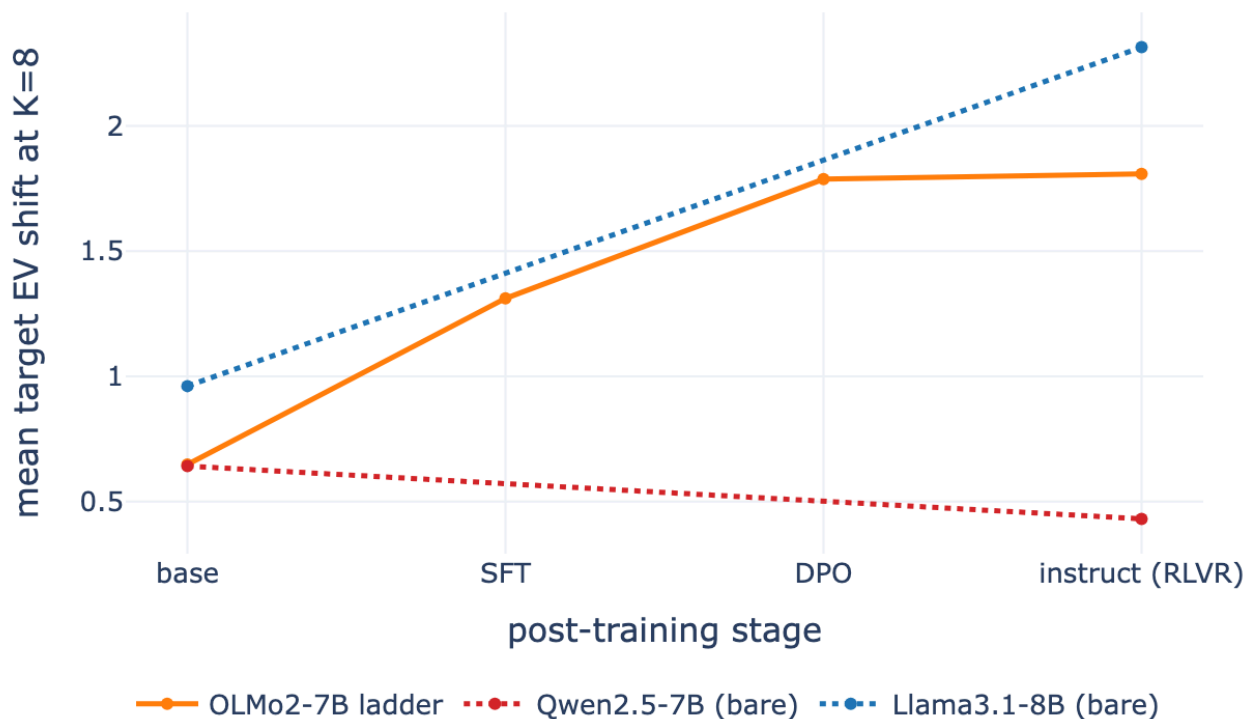


Figure 2: Post-Training Movement

Training Effects by Model:

- OLMo base→SFT: +0.66 [+0.26, +1.09]
- OLMo SFT→DPO: +0.48 [+0.19, +0.81]
- OLMo DPO→RLVR: +0.02 [-0.14, +0.20]
- Qwen base→instruct: -0.21 [-0.61, +0.13]
- Llama base→instruct: +1.35 [+0.52, +2.23]

Base models mostly don't move much for this intervention, and some don't move at all at K=8 (though Gemma base amusingly moves more for smaller doses than larger ones). Training mostly increases the movement, though Qwen remains flat. We only have OLMo for the individual training stages, but it looks like SFT and DPO both contribute. Gemma's behavior is particularly interesting here since it's one of the most responsive models once post-trained, but absolutely flat at K=8 in base.

But we should be skeptical about this table, in that base model identity is quite different from the identity installed by post-training. Looking at the model’s self-report on all the Saucier (2018) adjectives, one can see that base models are:

- uncertain at filling in self-reports (high entropy),
- vary their responses less than post-trained models (Qwen, a lot less), and
- to the extent that they do vary from the center, are more likely to just say yes to good things and no to bad ones, without further nuance. Removing the first principal component leaves very little left for Qwen.

Table 9: Base Models are Shapeless

model	mean EV	SD	H	r(sibling)	r(cohort)	r(PC1)	PC1-removed r	residual SD
Qwen2.5-7B-base	3.14	0.12	1.74	+0.53	+0.58	+0.58	+0.20	0.094
Llama3.1-8B-base	4.27	0.46	1.84	+0.44	+0.88	+0.80	+0.63	0.280
Gemma3-12B-base	4.24	0.46	1.89	+0.68	+0.86	+0.80	+0.55	0.271
OLMo2-7B-base	3.02	0.31	1.64	—	+0.38	+0.41	+0.05	0.286
Qwen2.5-7B	4.14	1.51	0.65	—	+0.93	+0.73	+0.86	1.023
Llama3.1-8B	4.16	0.52	0.72	—	+0.54	+0.24	+0.66	0.504
Phi4-3.8B	4.80	1.41	1.24	—	+0.94	+0.89	+0.76	0.649
cohort ref (n=11)	—	1.33	0.58	—	—	—	—	—

- 523-adjective self-report instrument, Likert scores.
- PC1 = the cohort evaluative axis (double-centered SVD over the 11 tuned cohort profiles).

Phi4 in the table above sits in an unusual state: it’s nearly as uncertain as a base model, but has the richer self-conception more typical of instruct ones.

Despite all these caveats, Llama3.1-8B-base maybe moved more than other bases ($p \sim 0.1$; n is only 20), so it’s possible that some of its plasticity is already in the base model.

3.4 Model Turns After Intervention

The models vary in their self-report; do they also vary in their actual behavior? Qwen2.5-7B and Llama3.1-8B were given the chance to answer a held out question after the K=32 treatment for each adjective, and each judged the other’s output. The results were in the same direction as the self report, although weaker:

- Llama’s responses were judged +1.56 Likert in the trait (10/20 were more than 1 point higher).
- Qwen’s, however, were judged +0.09 higher – that is, flat (0/20).

Caveat: this is only one question and rollout per adjective, so treat as directional, at best.

The mechanism that reinforces this shift in Llama or weakens it in Qwen is likely affecting how the models actually behave, either directly or first through the lens of self-perception.

3.5 Steering

The difference δ between the mid-layer activation on the residual stream at the end of the prefill between $K=0$ and $K=32$ can act as a sort of summary of the input. It's not small; the diff between $K=0$ and $K=32$ activations is 0.6-0.8 of the size of the $K=0$ activation.

This δ can be used to steer the model. Testing it on Llama3.1-8B and Qwen2.5-7B shows similar behavior to steering with the context: at $\alpha=2$, Llama shifts +1.85 (56% of $K=32$ shift), but Qwen only shifts +0.21 (38% of $K=32$).

Given Qwen's disinclination to update, its δ might be particularly non-vocalizable in the sense of ([Gurnee et al. 2026](#)) – in the kernel of J matrix. This turns out not to be so for Qwen, however: the portion of σ_δ^2 captured by the top right singular vectors is no smaller than expected for Qwen's activations.

4. Discussion

This note shows that there's evidence of a material difference in how some models act when shown manipulated samples of their own behavior. Some of these models are so responsive that it's easy to imagine that small differences in behavior could occasionally dramatically change the model's output.

So far, though, we don't have a strong mechanism that explains why some models update, while others don't. I had several guesses that didn't pan out:

- Base model differences are too small to explain it and sometimes point in the wrong direction, leaving out easy answers from the corpora.
- The identity-anchored system prompt doesn't change things much.
- A model's disavowal seems not to be particularly related to outcome.
- The model's representation of past stages isn't particularly near or far from its vocalizable representation.

There are, however, more things that can be tried and a few flaws to be fixed:

- There are 523 adjectives in the full set from ([Saucier 2018](#)), and human self-image data for all of them. Collecting data for all adjectives would allow us to connect it with human correlations, consider the effective dimensionality of the induced behavior, and narrow the confidence intervals.
- The longer $K=32$ rollouts should be run with enough user questions to make the user turns unique.
- The behavioral post-treatment rollouts should be run for all models with more user turns.
- The 120-token rollouts should be normalized back to 100 for those families.
- With more adjectives, it becomes viable to try to look into the patterns and compare disavowed vs not, *etc.*
- We can collect activations during the prefill and look for disbelief and roleplay detection in the j lens.
- *Some* adjectives move for Qwen; we can try to look for activation differences between those that do and those that don't, particularly with a bigger sample.
- I haven't investigated the activation differences between the uninstructed and instructed interventions.

- It would be interesting to rerun Soligo et al. (2026)’s experiment on this cohort to see if the same models were susceptible to their treatment. This might help point to a mechanism.

This particular intervention was fairly targeted, but one could imagine another version with self-perception updates intermittently between other turns as well. If some models are especially plastic, then multiple forms of later-turn analysis become more important, especially as Agents push the number of turns up.

5. Acknowledgments

This work was done with the assistance of Claude Fable 5, which had meaningful contributions to the experiment design, wrote most of the code, and ran the experiments. Other Claude models assisted in the larger project this was split off from, including some of the data this note borrows. This text is authored by the human listed above, and numbers in the text were checked against the primary outputs.

Thanks also to Ethan C. Brown for giving me the impetus to make this real and introducing me to psychometrics.

6. References

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